Individual Capstone Self-Assessment

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Part A - Individual Contribution

My senior design project, *PickyRTL,* ended with the creation a fully functional static analysis tool that automatically detects hardware Common Weakness Enumerations (CWEs) in Register-Transfer Level (RTL) code. The project started as a research co-op I began during the summer semester, and this year I took that work and built it into a complete tool that was accepted for publication at a peer-reviewed conference. My responsibilities encompassed every stage of the project from researching related work, designing detection algorithms, collecting a dataset of 39 HDL files, evaluating the tool, and writing the final research paper.

The skills I identified in my initial fall assessment proved directly applicable throughout the project, and in each step I built on top of them. My background in algorithms was essential for designing the AST traversal and CWE-specific detection logic, particularly the state-coverage, unreachable-state, and deadlock checks for CWE-1245, which required graph reasoning. My Technical Writing coursework paid off when writing a conference paper, where clarity in communicating my methods was just as important as the technical work itself. The emphasis on high-quality datasets and proper evaluation from my AI coursework shaped my approach to evaluating the tool. Rather than testing on a small or homogeneous sample, I deliberately assembled examples from multiple sources including the CWE MITRE website, HACK@DAC competition repositories, and the VUL-FSM dataset, totaling over 8,900 lines of code. One of the best accomplishments of the tool was achieving 100% precision and recall for CWE-1245 and CWE-1431 on the gathered dataset. These results demonstrated that static analysis works well for weaknesses that occur in structural. The most significant obstacle I faced was identifying security-sensitive registers for CWE-1233 and CWE-226 without full design context, which led to lower precision and recall for those CWEs. Working through that limitation deepened my understanding of the gap between rule-based static analysis and the richer semantic knowledge a human reviewer would bring to the same code.

Part B – Team and Collaborative Contributions

Although I was the sole student contributor on this project, the work was collaborative in the broader sense of research. I worked closely with my faculty advisor, Dr. Boyang Wang, throughout the year. He helped me identify which CWEs from MITRE's Most Important Hardware Weaknesses lists were both practically significant and amenable to rule-based static analysis. These scoping decisions were crucial to deliver an effective tool. His feedback on the research paper helped refine the content of the paper and help better explain the limitations of the tool. His expertise in the area and conducting research was extremely valuable throughout the whole year.

Working independently also reinforced lessons about motivating myself and coming up with my own ideas. With no teammates to divide responsibilities or to brainstorm ideas, the making of every decision fell on me. I had to manage my own timeline across the project, hold myself accountable to milestones without external reminders, and develop the discipline to recognize when an approach wasn't working and change course. Those experiences complement the collaborative skills I built during my co-ops, and together they have shaped me into a more well-rounded engineer. Overall, this project has improved my skills in many different areas including working as an individual while also collaborating with a team.